

# MAYUR ATHAVALE

Hyderabad, India • +91-7972833243 • mayat.dev1569@gmail.com

 linkedin

 github

 medium

 mayurathavale.com

 \_ ssh portfolio.mayurathavale.com

## SUMMARY

Software engineer with end-to-end production ownership across **distributed microservices, event-driven pipelines, and cloud infrastructure**. At Zotok.ai, built and operated systems spanning a **Go API gateway across 12 microservices**, a **multi-tenant auth layer** with strict input validation and secure coding practices, **AWS infrastructure via Terraform**, and an **agentic AI copilot** that drove measurable business impact. Comfortable across the full stack: backend systems design, cloud infrastructure, automated testing, and **React/TypeScript** frontends. Holds a high bar for correctness and operational discipline; comfortable debugging across multiple levels of the stack and shipping with confidence.

## EXPERIENCE

- **Software Development Engineer** **Jan 2025 – Present**  
*Zotok.ai* *Hyderabad, India*
  - Designed and owned a **multi-tenant identity and auth layer** (Hasura + NestJS proxy): dynamic schema generation per tenant, query guards, role-based access enforcement, and **strict input validation and secure coding practices** throughout; applied the same discipline to **React/TypeScript** frontend surfaces consuming these APIs end-to-end.
  - Built a **Go-based API gateway** routing traffic across **12 microservices** with auth middleware, rate limiting, and request routing via full **KrakenD configuration**; applied concurrency patterns and **performance tuning** to meet high-throughput SLAs reliably and safely.
  - Designed **event-driven async pipelines** on **SQS, SNS, DynamoDB Streams, and webhooks** modelled for **idempotency, reliability, and explicit failure-mode handling**; covered retry logic, dead-letter processing, and partial-failure recovery across async consumers with full test coverage.
  - Owned **AWS infrastructure end-to-end via Terraform** (ECS Fargate, RDS, SQS, SNS, DynamoDB, Secrets Manager, Elasti-cache/Valkey); built and maintained a **GitHub Actions CI/CD** pipeline with automated integration and functional tests as quality gates, cutting release cycles by **35%**. Implemented **CloudWatch metrics and structured logging** for production observability.
  - Delivered **zero-downtime database migrations** on production PostgreSQL (RDS) and significant **frontend bundle size reduction** via module federation refactoring, both planned and shipped with full test coverage and no production regressions. Authored **design docs and runbooks** covering architecture, failure modes, and operational procedures.
  - Architected an **agentic AI copilot** (FastAPI + LangGraph) with a **RAG pipeline** (OpenSearch KNN + BM25 hybrid retrieval) and 4 streaming strategies (SSE, AG-UI, LangGraph multi-mode); drove a **200%+ increase in daily sales leads**. Provisioned and managed **Kubernetes** workloads for service orchestration across the platform.

## PROJECTS & WRITING

- **PR Manager**  (Go, GitHub APIs, GitHub Actions) — CLI tool that automates end-to-end PR workflows: reviewer assignment, merge queuing, status checks, and cross-platform binary releases via **GitHub Actions CI/CD**. Built with plugin-style extensibility and DevOps tooling integration in mind.
- **Terminal Portfolio**  (Go, BubbleTea, Wish, Lipgloss, AWS EC2) — SSH-accessible TUI at `ssh portfolio.mayurathavale.com` with multi-tab navigation, SQLite visitor analytics, and live presence tracking. Deployed on **AWS EC2** with **GitHub Actions** hot-deploying via SCP and systemd, infrastructure managed end-to-end.
- **Technical Writing** — Published articles on system security, cloud infrastructure, and networking on Medium : [self-hosted VPN tunneling](#). 

## TECHNICAL SKILLS

**Languages:** Go, Python, TypeScript, JavaScript, SQL, Bash

**Backend:** REST APIs, Microservices, FastAPI, NestJS, Node.js, API Gateways (KrakenD), Event-driven Architecture (SQS, SNS, DynamoDB Streams), Streaming APIs, Webhooks

**Frontend:** React, Next.js 14, TypeScript, GraphQL, tRPC, Module Federation

**Databases & Caching:** PostgreSQL, MySQL, Redis, DynamoDB, OpenSearch, MongoDB, Firebase

**Infrastructure & DevOps:** AWS (ECS Fargate, EC2, RDS, SQS, SNS, DynamoDB, Lambda, Secrets Manager, Bedrock, CloudWatch), GCP, Terraform, Docker, **Kubernetes**, CI/CD (GitHub Actions), Caddy

**Testing & Quality:** Integration testing, functional/end-to-end testing, unit testing, quality gates in CI/CD, zero-downtime migration strategies, code review

**Observability:** CloudWatch, structured logging, metrics, distributed tracing

**Security:** Secure coding practices, input validation, role-based access control, secrets management

**Concepts:** System Design (LLD/HLD), Distributed Systems, Concurrency, Multi-Tenant Architecture, Resiliency Patterns, Performance Engineering, SOLID, Agile/SDLC

**AI/ML:** LangGraph, LangChain, RAG Pipelines, OpenSearch (KNN + BM25), Multi-Agent Orchestration

## EDUCATION

- **College of Engineering Pune (COEP)** **2020 – 2024**  
*B.Tech*